

4. Select system components using manufacturers' catalogues for process plants

Text Books

1. S.B. Tharkore and B.I Bhatt, Illustrated Process Equipment Design, Tata McGraw Hill (2000), 1st ed.
2. V. V. Mahajani & S.V. Umarji, Process Equipment Design, McMillan (2007), 4th ed.
3. S. Ray and G. Das, Process Equipment & Plant Design, Elsevier (2020), 1st ed.

Reference Books

1. IS 2825 (1969): Code for unfired pressure vessels
2. ASME B31.1 (2020): Power piping code
3. IS 1239 (2004): Steel tubes, tubular and wrought steel fittings

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessional (Assignments/Projects/Tutorials/Quiz)	25

UME631 : Solar Energy Engineering

L	T	P	Cr
3	0	0	3.0

Course Objective: The course intends to provide the fundamentals underlying solar energy utilization: Solar Thermal and Solar Photovoltaic. To impart the students, the ability to carry out heat transfer and optical analysis of these solar energy systems. To impart application based knowledge so that students are able to identify key parameters in solar energy utilization.

Syllabus

Introduction: Brief history of solar energy utilization - various approaches of utilizing solar energy, Economics, Energy life cycle analysis, Environmental considerations, Blackbody radiation approximation of solar energy-Planck's Distribution Law, Wien displacement law, Stefan - Boltzmann law. Photoelectric effect-Einstein's theory of photons.

Solar Radiation: Extra-terrestrial and terrestrial insolation, instruments used for measuring solar radiation, empirical correlation for predicting available solar radiation, computation of solar radiation on horizontal and tilted surfaces.

Flat Plate Collectors: Selective surfaces- materials- optical and radiative properties, construction details, heat transfer analysis, estimation of losses, collector efficiency and standard testing procedures.

Concentrating Collectors: Constructional details of various concentrating collectors- parabolic trough collectors, compound parabolic collector, paraboloid dish collectors, and central receiver collector, Designing concentrators and heat collector elements for achieving high optical and thermal efficiency, heat transfer analysis, estimation of losses, collector efficiency and standard testing procedures.

Evacuated Tube Collector (ETC): Description and working principle of ETC systems. construction details of ETC, selection and installation of ETC systems, performance parameter tests of ETC systems.

Storage of Solar Energy: Types of Energy Storage - Thermal Storage - Electrical Storage - Chemical Storage - hydro-storage.

Photovoltaic Basics: Structure and working of Solar Cells - Types, Electrical properties and behavior of Solar Cells – Cell properties and design.

Solar Power Plants: Solar Thermal Power Plants- Solar Power cycles (Vapour cycles, Organic cycles, Combined cycles, Binary Cycles. Stirling and other cycles) - basic construction and working. Photovoltaic Power Systems - System Integration - Energy Storage, PV Cell Interconnection and Module Fabrication - PV Modules and arrays - Basics of Load Estimation. **Other Solar Thermal Applications:** Solar refrigeration and air-conditioning, solar pond, solar desalination.

Research Assignment: Students in groups shall choose one topic of their interest relevant to solar energy utilization. Each group shall be required to submit a term paper relevant to the same. A term paper shall essentially be original work discussing a topic in detail- new design/modification proposed and the supporting analysis. Each group shall be required to submit the completed term paper at the end of the semester.